Quantitative Aptitude & Speed Math Formulas for Placements & CAT

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1. SPEED MATH & PERCENTAGE MULTIPLIERS

• **Fraction to Percentage Equivalents:** 1/6 = 16.67%, 1/7 = 14.28%, 1/8 = 12.5%, 1/9 = 11.11%, 1/11 = 9.09%, 1/12 = 8.33%, 1/13 = 7.69%, 1/14 = 7.14%. Essential for mental arithmetic.
• **Successive Percentage Formula:** Total change = (A + B + (A * B)/100)%. Example: 20% increase followed by 10% decrease = (20 - 10 - 2) = +8% net gain.

2. TIME, SPEED & DISTANCE

• **Average Speed (Equal Distances):** Avg Speed = (2 * S1 * S2) / (S1 + S2). Harmonious mean of the two speeds.
• **Relative Speed:** Same direction: S_rel = |S1 - S2|. Opposite direction: S_rel = S1 + S2. Used for train crossing problems and circular track chases.

3. PERMUTATIONS, COMBINATIONS & PROBABILITY

• **P&C Fundamental Rules:** nPr = n! / (n - r)! (order matters, arrangements). nCr = n! / (r! * (n - r)!) (order does not matter, selections). Key identity: nCr = nC(n-r).
• **Bayes Theorem:** P(A|B) = [P(B|A) * P(A)] / P(B). Calculates posterior probability of event A given prior observation B.